

Sylvia Mesicek

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EDUCATION

University of Utah

Honors Bachelors of Science in Physics & Mathematics

Minor: Astronomy

Cumulative GPA: 4.00 / 4.00

Salt Lake City, Utah

Aug. 2022 - Present

Salt Lake Center for Science Education

High School Diploma

Cumulative GPA: 4.00 / 4.00

Salt Lake City, Utah

Aug. 2019 - June 2022

RESEARCH

Critical Collapse of Black Holes with Finite Difference Methods

Sep. 2022 - Present

- Developed advanced codebase (20,000 lines of code) from scratch for numerically solving PDEs.
- Derived a novel hyperbolic formulation of the Z_4 extended Einstein Field Equations in axisymmetry.
- Designed and implemented innovative methods for adaptively refining domains using interpolating wavelets.

Advisor: **Dr. John Belz**

X-ray Spectral Analysis of AGN in the NEP field

May. 2025 - Present

- Performed x-ray spectroscopic analysis on XMM-Newton observations of 62 selected AGN in the JWST-North Ecliptic Pole field.
- Used Bayesian analysis and frequentist best-fit models to measure and constrain N_H values of these AGN.
- Supplemented prior work studying obscured AGN in the NEP field.

Advisor: **Dr. Dan Wik**

Formal Verification of Operating Systems

Sep. 2022 - May 2024

- Verified correctness of low-level OS code by building proofs with Dafny and Verus.
- Built formally verified abstractions for memory allocators and linked lists.
- Coauthored paper on work presented at SOSP 2023, a leading conference in operating systems programming.

Advisor: **Dr. Anton Burtsev**

PUBLICATIONS

- **Atmosphere: Towards Practical Verified Kernels in Rust.** Xiangdong C., Zhaofeng L., Mesicek L.¹, Narayanan V., and Burtsev A. Published in *KISV '23: Proceedings of the 1st Workshop on Kernel Isolation, Safety and Verification*.

¹Citation under previous name.

PRESENTATIONS

- **Simulating Black Hole Collapse from Axisymmetric Scalar Fields using Modern Finite Difference Techniques.**

American Physical Society (4 Corners Conference) - Logan, Utah. Oct. 2023

L. Mesicek¹, Sean Johnson, John Belz.

- **Axisymmetric Critical Phenomena using High Order Finite Difference Methods.**

University of Utah Undergraduate Research Symposium - Salt Lake City, Utah. Aug. 2023

L. Mesicek¹, Sean Johnson, John Belz.

TECHNICAL SKILLS

Languages: C/C++, Python, Rust, Julia, Fortran

Libraries: NumPy, SciPy, Pandas, Matplotlib, AstroPy, SymPy

Miscellaneous: LaTeX, ParaView, Xspec

ADVANCED COURSEWORK

- **Stars & Stellar Populations.** Graduate level course in stellar astronomy and physics, covering stellar structure, formation, energy transport, nucleosynthetic reactions, energy production, and stellar ensembles. *Designation: ASTR 5560, Grade: A*
- **Cosmology.** Graduate level course in modern cosmology, covering dark matter, dark energy, cosmological density fields, growth of density perturbation, inflation, anisotropy in the cosmic microwave background, galaxy formation, and cosmic reionization. *Designation: ASTR 5580, Grade: In Progress*
- **General Relativity.** Graduate level course in general relativity, covering manifolds, differential geometry, the Einstein Field Equations, the Schwarzschild solution, the Kerr solution, and gravitational radiation. *Designation: PHYS 7720, Grade: A*
- **Electrodynamics and Special Relativity.** Graduate level course in electrodynamics and special relativity, covering the tensorial formulation of electromagnetic fields, Lagrangian formulation of relativistic mechanics, electrostatics, magnetostatics, and multipole expansions. *Designation: PHYS 7110, Grade: A*
- **Computational & Statistical Methods.** Graduate level course in statistical methods, covering stochastic process simulations, Monte Carlo methods, Bayesian analysis, and machine learning algorithms. *Designation: PHYS 7730, Grade: A*
- **Analysis of Numerical Methods.** Graduate level course in numerical analysis, focusing on solving PDEs numerically. Topics covered included Runge-Kutta methods, multistage methods, interpolation, finite difference approximations, and continuous galerkin spectral methods. *Designation: MATH 6620, Grade: A*
- **Classical Mechanics.** Undergraduate course in classical mechanics, covering advanced Newtonian mechanics, Lagrangian mechanics, Hamiltonian mechanics, rigid-bodies, and central-force problems. *Designation: PHYS 4410, Grade: A*
- **Quantum Mechanics.** Undergraduate course in quantum mechanics, covering the harmonic oscillator, free particles, scattering, three dimensional quantum mechanics, the hydrogen atom, spin, and Dirac notation. *Designation: PHYS 5450, Grade: A*
- **Computational Physics.** Undergraduate course in computational methods for physics, including NumPy and Matplotlib, statistical methods, data fitting, numerically integrating ODEs, and extracting waves via fourier transforms. *Designation: PHYS 3730, Grade: A*
- **Thermodynamics & Statistical Mechanics.** Undergraduate course on thermodynamics and statistical mechanics, covering the laws of thermodynamics, entropy, temperature, heat engines, refrigerators, phase transitions, Boltzmann statistics, Bose-Einstein statistics, and Fermi-Dirac statistics. *Designation: PHYS 3760, Grade: A*

- **Observational Astronomy.** Undergraduate lab in observational astronomy, covering the operation of telescopes and cameras, image processing, and data analysis through code. *Designation: PHYS 4060, Grade: A*

AWARDS AND RECOGNITION

Barry Goldwater Scholarship	2025
<i>Awarded by the Barry Goldwater Scholarship and Excellence in Education Foundation for outstanding career and research potential in physics and astronomy.</i>	
Walter W. Wada Endowed Scholarship	2025
<i>Awarded by the University of Utah's Department of Physics and Astronomy to an outstanding undergraduate student.</i>	
Michael Zhao Memorial Scholarship	2025
<i>Awarded by the University of Utah's Department of Mathematics to an outstanding undergraduate student.</i>	
James B. & Betty Debenham Scholarship	2024
<i>Awarded by the University of Utah's Honors College for outstanding student involvement and achievement on the path to an Honors Degree.</i>	
University Opportunity Research Program	2024
<i>Awarded by the University of Utah's Office of Undergraduate Research to fund my work with Dr. Belz in the spring and summer of 2024.</i>	
College of Science Dean's Scholarship	2023, 2024, 2025
<i>Awarded by the University of Utah's College of Science for outstanding undergraduate academic achievement in science classes.</i>	
Summer Undergraduate Research Fellowship	2023
<i>Awarded by the University of Utah's Department of Physics and Astronomy for academic merit and research experience to fund my work with Dr. Belz over the summer of 2023.</i>	
Sweet Candy Scholarship	2023
<i>Awarded by the University of Utah's Honors College for outstanding student involvement and achievement on the path to an Honors Degree.</i>	
Physics and Astronomy Recognition of Excellence	2022 & 2023
<i>Awarded by the University of Utah's Physics and Astronomy for outstanding undergraduate academic achievement in physics classes.</i>	
University of Utah Flagship Scholarship	2022 - 2026
<i>A merit scholarship awarded by the University of Utah to incoming freshman for academic achievement in high school.</i>	